

Article

Electrification Planning for Off-Grid Communities in Sub-Saharan Africa: Advancing Energy Access

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Abstract: Sub-Saharan Africa, especially its rural areas, faces significant challenges in achieving universal electrification despite its abundant renewable energy resources. The region has the highest population without access to electricity, largely due to economic, infrastructural, and geographical barriers. Energy poverty is a critical issue that hinders sustainable development and exacerbates inequalities. Namibia's sustainable energy policy aligns with the global Sustainable Development Goals (SDGs), particularly SDG 7, which aims to provide affordable and reliable modern energy access for all. The policy emphasizes mini-grids and decentralized power systems as key strategies for rural electrification. However, despite increased deployment of mini-grids, these solutions often struggle with long-term sustainability. This research explores cost-effective electrification strategies through scenario-based modeling to reduce energy poverty and expand energy access in Namibia's rural communities, focusing on the existing mini-grids in Tsumkwe and Gam. Using a comprehensive methodology that incorporates HOMER Pro for mini-grid capacity expansion and MS Excel for evaluating main-grid extensions, this study aims to identify the most feasible and economical electrification solutions. The analysis compares electricity supply, total net present cost, and the levelized cost of electricity across these systems. The findings will offer insights into addressing energy poverty in Namibia and provide recommendations for sustainable and scalable rural electrification across Sub-Saharan Africa.

Keywords: off-grid communities; grid extension; mini-grid expansion; rural electrification; Sub-Saharan Africa; Namibia



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1. Introduction

1.1. Energy Situation in Sub-Saharan Africa

Access to reliable and affordable electricity is [1] a critical challenge in Sub-Saharan Africa (SSA), where almost half the population remains without access to electricity [2]. In 2022, only 51.4% out of a total of 1.21 billion had access to electricity. This limited access to electricity is further intensified as regions get more remote and rural. An electricity access rate of 80.7% [3] versus 30.7% [4] for urban and rural areas, respectively, was recorded in 2022, emphasizing the disparities between these communities. While urban areas have, in most SSA countries, not reached universal access, the high concentration and density of the national electricity grids set the access rates quite high [5].

Urban consumers are therefore normally connected to the national grids due to their proximity to established distribution systems. This does not hold true for most rural and remote areas. As the distance increases from grid-concentrated regions, the costs for grid extension (i.e., including transmission and distribution) increase [6,7]. This fact discourages grid extension to rural areas, which make up about 55% of the total SSA population [8]. In 2019, 85% of urban areas compared to 35% of rural areas of SSA were grid connected [9], which solidifies the infeasibility of centralized grid extension as a rural electrification solution. It must be noted that approximately half a billion people reside in these rural areas [10]. This group is then therefore disadvantaged since the economic growth and

social development are also hampered [11]. It is, for this reason, that this paper specifically focuses on rural areas of SSA to navigate means to enhance rural electrification, which positively affects other socio-economic standings.

SSA's countries continue to work towards achieving the Sustainable Development Goals (SDGs), especially goal 7, which calls to 'ensure access to affordable, reliable, sustainable and modern energy for all' [12,13]. Among the efforts to bridge the electrification challenge in rural areas are the policies and frameworks that have been formulated in SSA member states and unions [14], emphasizing the adaptability of renewables. Ranging from large-scale projects like wind farms, biogas plants, hydropower plants, and solar mini-grids to smaller projects like containerized photovoltaic (PV) systems and solar home systems (SHSs), a number of renewable energy projects have been established [15–18]. However, for the electrification of communities, larger projects such as solar mini-grids are preferred over small and sometimes single-user-specific solutions such as stand-alone containerized and solar home systems [19–21].

While all these electrification projects seek to increase electricity access, a core factor to consider is the sustainability of the applied approach. In fact, it has been argued that solutions such as solar mini-grids are much more sustainable compared to solar home systems in rural areas [22]. For SSA's rural electrification specifically, and due to the abundance of solar resources, falling costs of solar panels, and the fast technological advancement in the solar industry, solar-based solutions gained more traction [23,24]. Solar mini-grids, containerized systems, and SHSs have increasingly been implemented [16,25]. However, the discussion around grid extension has not ceased due to the success rate of grid performance in urban areas compelling rural residents preferring the same solution [26,27].

Solar mini-grids are used to supply electricity to rural communities [16], herein referred to as off-grid communities, as they present a cost-effective [28] solution unlike grid extension. This approach in relation to its sustainability (i.e., regulatory, e.g., licensing and operation approvals; financial, e.g., high investment costs; technical, e.g., the lack of on-ground technical experts; and operational, e.g., long distances to travel for troubleshooting system components) also faces challenges and therefore fails to supply continuous electricity to consumers [29–31]. As off-grid communities get electrified, their electricity demand increases due to factors such as residents acquiring new electrical appliances and business establishments. Once these factors lead to community electricity demand surpassing the generation capacity, electricity blackouts become inevitable, and operators put in place demand-side management strategies [32]. Off-grid communities, therefore, while electrified, still hit a roadblock that requires, mainly, technical intervention. This paper therefore solely focuses on these established communities and strategies to eradicate energy poverty. Solar mini-grids are the technology discussed due to the justification that they are not only preferred but also more sustainable than solar home systems and other small-scale solutions.

Since both grid extension and solar-based mini-grids already exist either in rural areas or in their proximity, the need to assess the feasibility of these two solutions in addressing off-grid communities' electrification challenges is key to identifying which solution would attain sustainable off-grid electrification. SSA records a significant surge of implemented mini-grids, which necessitates the analysis of their performance and sustainability in the long run relative to the addressed communities. Moreover, grid extension is in most SSA countries still within the plan of the electrification strategies even for rural areas [33,34]. While this is the case, the literature also suggests a lot of grid extension challenges, such as grid instability and high investment costs, thus are often recommended to areas close to existing power grids [35].

The next section looks at the current off-grid electrification strategies with a focus on solar mini-grids, grid extension, and the electrification status quo in SSA as a basis for the analysis performed in this paper.

1.2. Off-Grid Electrification Status in Sub-Saharan Africa

As mentioned in the previous section, renewable energies are at the forefront of rural electrification in SSA [15,17,36,37]. Though they vary in terms of capacities depending on the geographical location, available capital and investment fund, and the technical power system design factors, it is safe to say that SSA has concentrated its efforts mainly on solar-based solutions [25,38]. Mini-grids are seen to be implemented by both private investors and governments [39], as is the case in Namibia, for instance. The number of solar mini-grids has increased dramatically in SSA from an estimated 500 that were installed in 2010 to more than 3000 in 2023, accounting for more than a 500% increase in two decades. From 2010, mini-grid connections increased to 11 million, resulting in a rise in SSA's rural electrification rate from 17% to 28% within a decade [15].

In Nigeria and Kenya, for example, solar mini-grids have been prioritized to not only electrify rural areas but also inject electricity into the national grid. In Senegal, solar mini-grids have been identified as the second preferred solution, after grid extension, for rural electrification [40]. Kidenda [41] states that settlements outside a one-kilometer radius from an existing grid should be electrified by solar mini-grids. In addition, he states that solar mini-grids should be the frontier strategy in rural areas with population densities below 1000 people per km².

Regardless of the consideration for solar mini-grids, grid extension continues to be a strong contender in rural electrification. In Nigeria, grid extension is the top preferred rural electrification strategy. In Namibia, grid extension is also an ongoing process, even though extension plans remain vague [42].

In the broader context of rural electrification in SSA, most of the member countries consider several approaches to reach the SDG7 targets. However, grid extension and solar mini-grids are at the forefront [10,33,34]. Similarly, policies have been put in place advocating for the same across many SSA member states, even though they could be strengthened [43]. Policies could highlight the security of mini-grid operators upon the arrival of the grid to discourage the implementation of solar mini-grids in the first place [15]. This becomes necessary as grid extension decisions have been linked to political decisions and thus remain very opaque and unpredictable [44].

The dominance of solar mini-grids and grid extension makes it crucial to identify their role in complementing off-grid communities and determine scenarios to enhance the existing systems. This paper uses a case of Namibia, which has already established off-grid communities for over a decade now.

1.3. Introducing the Case of Namibia

Approximately 46% [45] of Namibia's population resides in rural areas where access to electricity is limited. Namibia's electricity generation relies on a mix of petroleum, hydropower, wind, solar PV, imported electricity, and coal [46]. The country heavily relies on imports, especially from South Africa, Zambia, and Zimbabwe, which further complicates its energy security [47]. It is significant to note that this is due to the country's low local generation capacities. As of 2022, hydropower was the highest-explored renewable energy source. Aili et al. (2022) [42] mention that the country's peak demand is met by a local-to-import ratio of 40:60. Evidently, this highlights the need for an increase in in-country self-generation. The authors add on by noting that a deployment of 278 MW of renewable capacities was planned for the next years. The government of Namibia (GoN) not only expresses its efforts to increase its generation capacity but also specifically identifies the role of renewables in reaching the targets as defined in SDG7. The GoN has set out a target to electrify all Namibian households by 2040 by means of either grid extension or off-grid electrification [33]. Additionally, with a renewable energy policy put in place, the GoN intends to optimize and further incorporate renewable energies in the electrical energy mix [48]. The set goal for Namibia was to rise from 27% in 2015 to 70% by 2030 [42]. It also states the role of solar photovoltaics, especially in mini-grid and stand-alone applications

in electrifying rural settlements without connection to the utility grid and stand-alone systems, such as solar home systems for individual uses and applications.

Up until 2022, the solar-based power production in Namibia was approximately 174.5 MW. Solar power production, amongst other renewables, follows hydropower in terms of the installed capacities in Namibia [42]. This is supported by an abundance of and high solar irradiation values of about 2447 kWh/m² [49,50], which presents an opportunity to harness solar power for the electrification of unelectrified regions. Likewise, this also highlights room for improvement to ensure the installed mini-grids, which have adequately met the need for rural electrification in Namibia, so as to attain a sustainable electricity supply as is advocated for by SDG7.

In 2022, only 55.23% of the population had access to electricity, where urban and rural access was 74.74% and 33.21%, respectively [51]. This urban–rural electricity access disparity highlights the challenges Namibia faces in providing electricity to remote areas. Namibia faces significant rural electrification challenges due to its low population density (i.e., 3 people/km²) [33], high costs of grid extension [50], and high tariffs [52]. Grid extension has, in some areas, been opted for as the most fitting electrification solution in Namibia. For instance, the Mangetti Dune region, which is the closest grid point to Tsumkwe and Gam, was electrified through grid extension. However, due to long extension distances over which low- to no-electricity consumers reside, high costs were incurred; meanwhile, the electricity available to the end consumers is very unstable. Through a site visit, this study identified the single low-voltage wire extended over hundreds of kilometers and a low-voltage transformer (Figure 1, i.e., 200 kVA). The Namibian Broadcasting Corporation records [53], ‘CENORED are tasked to construct the electricity power lines from Maroelaboom through Mkata village to Mangetti Dune for now’ and adds on, ‘The project is expected to cost over N\$40 million for approximately 125 kilometres from the Maroelaboom Police Station near Grootfontein’. However, the government still allocates a high budget for subsidizing electricity costs, proof of high grid extension investment costs [54]. Low population densities in Namibian settlements limit grid extension due to the high need for subsidies. Moreover, the extended grid, as is the case in Mangetti Dune, has proven to not offer stable and sustainable electricity solutions.



Figure 1. Mangetti Dune transformer.

The GoN, through its Ministry of Mines and Energy (MME), however, established, among other solar-based solutions, mini-grids in rural areas. Of Namibia’s three established off-grid communities equipped with mini-grids are Tsumkwe and Gam, which are also the

largest. The presence of such solar-based off-grid communities highlights the potential that the GoN has identified in solar technology to address the country's electricity access challenges. Likewise, these systems have displayed operational challenges limiting electricity availability to the end consumers. The systems, which are approximately a decade old, are currently failing to meet the communities' electricity demands due to community growth (i.e., including economic activities, population growth through in-migration, and increase in energy-intensive appliance ownerships). In 2022, it was noted that the mini-grid had a surge in diesel consumption as a result of increased consumption and the presence of batteries that had already reached their end of operation lifetime [55].

Namibia's remote regions set a great opportunity to study the feasibility of utilizing the two community-based electrification strategies in enhancing off-grid electrification. The availability of challenges for both scenarios underscores the urgent need to study reliable and sustainable electrification solutions in Namibia's existing off-grid communities.

1.4. Aim of the Paper

Despite the increased number of implemented mini-grids and other off-grid decentralized electrification solutions in off-grid communities, electricity availability challenges continue to persist. Statistics continue to show increasing access rates and the number of implemented mini-grids; however, they fail to capture the actual number of operational mini-grids. The necessity to monitor mini-grid performance goes hand in hand with monitoring the demand growth in the respective off-grid communities.

This paper uses the case of Namibia's two largest off-grid communities and the associated challenges of the installed mini-grids in providing uninterrupted electricity to the end-users. Because the main challenge arises from community electricity demand exceeding the mini-grids' generation capacity, this paper also looks at studying the generation and consumption trends before analyzing the enhancement scenarios. By using the basis of the electrification plan set out by the GoN to use either grid extension or off-grid electrification, this paper also considers these two scenarios as electrification enhancement strategies. While there are other communities where stand-alone solar systems, such as containerized systems in Utsathima, and solar systems specifically for one consumer, such as is the case at the Mangetti Dune health center, are implemented, this paper focuses on solar mini-grids, which are implemented to serve communities. These solar mini-grids are therefore directly affected by the increase in demand by the community.

The objective is to analyze the viability of expanding mini-grids or extending the existing grid to increase generation capacity and reduce energy poverty in off-grid communities.

1.5. State of the Art and Research Gap

To fully understand the contribution of this paper's study, a literature review on previous studies and the status quo of the discussions around electrification enhancement and rural electrification was performed. The most relevant studies are highlighted below.

Lefu et al. [56] point out the factors that account for the predominance of grid extension regardless of the tremendous amounts of research justifying the feasibility of decentralized rural electrification approaches. The availability of many funding schemes for subsidizing grid electricity costs surpasses the schemes to fund off-grid electrification solutions, which oftentimes are privately operated. However, their study also shows that subsidies required for off-grid electrification solutions would remain lower than those for grid extension. The study highlights the potential for solutions such as solar mini-grids but does not in detail navigate the technical configuration of these off-grid decentralized electrification solutions.

Cyril et al. [38] navigate the most relevant and recent research on the topic of rural electrification in SSA, putting together the literature from a timespan of up to 10 years ago. Their study highlights the potential of solar mini-grids to electrify rural settlements, not ignoring that they come with economic challenges and investment risks. However, the advantages that they offer are reliability and high-quality electricity supply while using local and natural resources, with the possibility for grid integration upon grid arrival. On

the one hand, they point out that grid extension is only feasible for rural areas within defined proximity with an existing grid. The authors insist that solar mini-grids require financing, regulatory frameworks, and technical capacities to facilitate their feasibility. The paper emphasizes these findings but fails to present site-specific findings based on data or site surveys. In addition, this paper only focuses on electrification strategies without putting forward enhancement scenarios for existing solar mini-grids.

Babayomi et al. [15] study renewable mini-grids deployed across SSA and identify the dominance of solar energy in this regard. They rephrase the importance of deploying renewable energy mini-grids in bridging the electrification gap between SSA and the rest of the world. They go further to present design methods considering community needs assessment, which they present to be very low in remote rural areas in comparison to urban areas. While they acknowledge tax exemptions on some solar components, they also present the bureaucratic inefficiencies within the implementation of renewable mini-grids, such as in license securing and approvals to operate mini-grids. Their paper also studies policies in several SSA countries, which mostly align with the interest to deploy mini-grids to increase rural electricity access. The paper, however, fails to present technically and economically feasible configurations to enhance existing mini-grids in addressing low rural electricity access rates.

Samatar et al. [57] use a case of a rural district in Somalia to assess the feasibility of renewable microgrids. Using different configurations, the techno-economic analysis and economic impact assessment were carried out to determine sustainable energy supply. The results showed solar PV/wind/diesel generators and battery energy storage systems to offer an optimal electricity supply for the district. While the paper is specific to Somalia and presents first steps to follow in determining the most optimal power system configurations, the paper fails to compare the findings to grid extension.

The literature is saturated with papers analyzing the feasibility of different renewable technologies in the efforts to increase electricity access in rural SSA. While different configurations are tested, research emphasizes utilizing solar energy in SSA due to the abundance of it. However, it is also pointed out that for these solutions to sustain, policies and regulatory frameworks, financing schemes, and technical capabilities are crucial [58–61]. On the other hand, there is a lack of literature on enhancing the electrification of existing and established off-grid communities. In addition, very limited literature is available in the context of SSA, and many existing studies use cases from Asia.

The methodology used in assessing the feasibility of the electricity enhancement scenarios is also another area of focus. Within this study, a literature review was conducted to identify previous approaches.

Yibeltal et al. [62] examined capacity expansion planning for off-grid PV mini-grids in rural Africa, focusing on developing optimal long-term strategies to meet evolving electricity demand in remote communities. They evaluated capacity expansion of the mini-grid system with a range of maximum annual capacity shortages using a multi-year optimization tool in HOMER Pro Version 3.14.5 software, aiming to improve the reliability and sustainability of the system. However, they did not compare it with other electrification scenarios, such as national grid extension. For the maximum annual capacity shortage values above 0%, the authors did not consider the use of an alternative option for the unmet load demand.

Moner et al. [63] compared the costs of grid extension and PV/hybrid mini-grid systems for electrifying rural SSA. They developed a formula to calculate and compare the LCOE for both options, factoring in distance from the main grid, connection costs, and community electricity demand. They also calculated the carbon emissions from the diesel generator to cover the same electricity demand as the mini-grid does. Moreover, authors only calculated LCOE for the comparison and no other key performance indicators (KPIs) like break-even grid extension distance.

Abdelhamid et al. [64] analyzed the feasibility of implementing remote PV mini-grid systems in five communities in Chad using a techno-economic approach. They examined

system configurations, capital and operational costs, and the levelized cost of electricity (LCOE) to assess economic viability. The study also explored financing methods and business models suitable for Chad's energy sector, providing insights into the integration of technical expertise and economic analysis for mini-grid electrification in remote areas. Moreover, the findings were not generalized to all localities in Chad, and the potential of hybrid photovoltaics was not explored.

Szabo et al. [65] compared the electrification costs of distributed solar, diesel generation, and grid extension in rural Africa. They analyze initial investments, operating expenses, and life-cycle costs. In their paper, the authors based their analysis on the assumption that one-third of the energy is consumed during the daytime and two-thirds during the evening and night, rather than using an hourly time series. Additionally, carbon emissions were not considered in the comparison for the diesel generator with the other two options.

The above research studies both show a lack of comparative study and focus either on one electrification strategy, electrification enhancement strategy, or a limited range of KPIs, such as only LCOE. There is, therefore, a need to analyze the viability of the two most common electrification strategies in the context of Namibia, that is, grid extension and mini-grid expansion as enhancing approaches for existing mini-grids. Incorporating a broader range of KPIs to assess this is crucial, which is why this paper uses the levelized cost of electricity, total net present cost, carbon emissions, and even the grid extension break-even distance as the main KPIs to determine the most effective electrification enhancement strategy.

While previous studies thoroughly evaluated off-grid electrification strategies and suitable renewable energy technologies, there is a lack of studies focusing on existing mini-grids. The evaluation of hybrid PV mini-grid systems has not considered environmental impacts such as the carbon emissions from the use of diesel generators. In addition, there is limited consideration of alternative solutions for unmet load demands, while key performance indicators beyond the LCOE, such as break-even distances for grid extensions, are also often overlooked. This paper not only looks at an understudied concept but also applies a holistic methodology that considers and presents results across the technological, economical, and environmental fields.

2. Materials and Methods

To meet the aim of the paper, this section outlines the methodological approach used, including the used software.

The research methodology followed in this paper is presented in Figure 2. It outlines the systematic approach used to assess the feasibility of the analyzed electrification enhancement solutions for Tsumkwe and Gam.

The first step analyzes the existing mini-grid systems in both communities to understand the current operating scenario. To do so, measurements and recorded data were used to analyze system performance in terms of electricity production, but also on the consumer side, the consumption patterns were studied. The measurement and recorded data were available from 2017 to late 2022; however, the years 2021 and 2019 were mainly used as the basis for this paper, as they present complete sets of data. In addition, the year 2020 was eliminated due to the influence of the COVID-19 pandemic, which therefore fails to reflect traditional consumption and production patterns. Among the data collection methods were mini-grid monitoring computers, which record the solar generation, diesel generation, and consumption at a one-minute resolution. Based on this data, an analysis of the annual energy generation and consumption profiles of mini-grids was performed. The objective of the first third of the methodology is to establish the electricity capacity shortage as generated by the existing mini-grid infrastructure and thus, in the next step, analyze the feasibility of the herein-studied electrification enhancement strategies. At this first step, diesel consumption and carbon emissions are also calculated as per the summed-up annual values for each of the two off-grid communities. The calculated values are also supporting inputs for the performed economic analysis of mini-grid components to calculate the total net present cost (TNPC) and the levelized cost of electricity (LCOE).

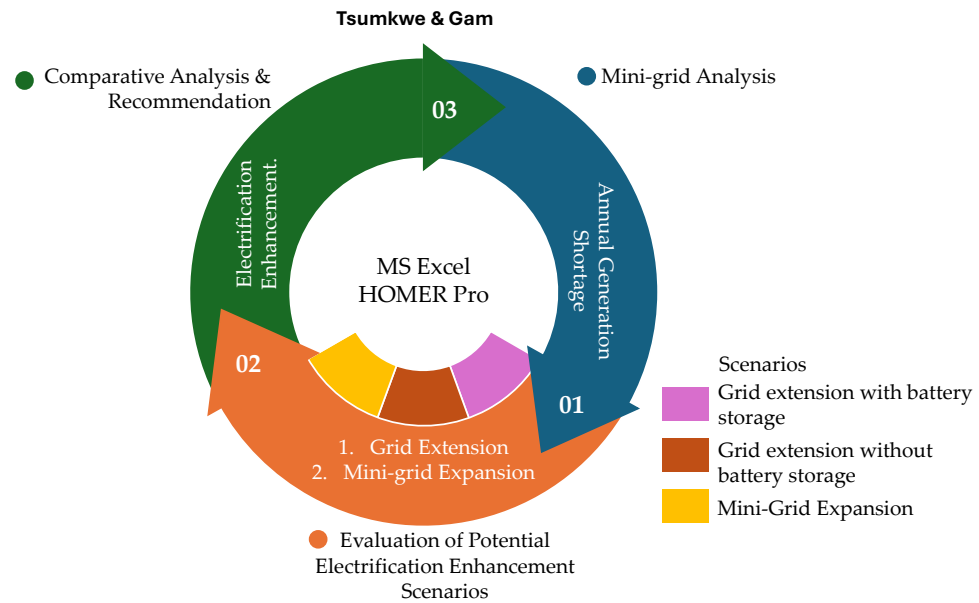


Figure 2. Research methodology.

In the second step, the two derived main scenarios for off-grid electrification enhancement are evaluated for both Tsumkwe and Gam. As shown in the figure, this step addresses the capacity shortage identified in the first step, aiming to achieve uninterrupted electricity supply, as discussed in later sections. The selection of these two enhancement scenarios (i.e., grid extension and mini-grid capacity expansion) also stems from the discussion within Tsumkwe and Gam and amongst Namibian energy stakeholders on whether they are feasible options to complement the now undersized mini-grids. As a matter of fact, the national grid is only 82 km away from Tsumkwe and 143 km from Gam, and therefore questions arise on whether integrating the grid to the mini-grids may enhance electrification and finally solve the energy poverty challenge. Therefore, from a technical point of view, this paper seeks to analyze the feasibility of expanding the mini-grid generation capacities through those two in-reach potential scenarios. The evaluation was performed through MS Excel Version 2409 Build 18025.20214 and HOMER Pro. The former was used to analyze the electricity consumption and generation values, while the latter was used to find optimal system configurations for the scenarios developed within this methodology.

In the case of the national grid extension, MS Excel is utilized to calculate the techno-economic performance.

To thoroughly evaluate the feasibility of grid extension, the following two categories are considered: 1. grid extension considering existing mini-grid battery storage, and 2. grid extension without battery storage. The two sub-categories differ in that one assumes that the existing mini-grid-installed battery capacity is fully functional and is useful energy storage, while the second assumes otherwise. The justification for this is that the installed batteries at the mini-grids are beyond their lifetime, and therefore a realistic analysis is required. In design, this affects the electricity supply and reliance on the main grid. To calculate the electricity supply from the grid in the first sub-category, an analysis in MS Excel was performed considering that if there is PV production, the electricity supply priority is for the immediate demand from the communities; any surplus is then fed into the batteries installed on site. At the bottom of the priority chain is feeding electricity into the grid, only when the batteries are fully charged and the immediate electricity demand from the communities is less than the generated electricity. In the second category, the national grid takes on the role of a backup generator. It therefore presents a case covering the nighttime demand and instances during the day when there is no solar production or the mini-grid is unable to meet the immediate demand. The surplus-generated electricity production from PV modules during the day is fed into the main grid.

The economic analysis was performed to understand the financial feasibility of electrifying the region by extending the national grid from the nearest accessible grid to the mini-grid site. The KPIs to evaluate the financial feasibility are LCOE and TNPC. To calculate these KPIs, the distance from the last grid point to Tsumkwe and Gam, the per km capital cost of grid extension, operation and maintenance costs, and the electricity demand of the community were taken into consideration. These values were obtained through a thorough literature review.

For scenario 2, namely, mini-grid capacity expansion, optimization is performed in HOMER Pro software. The methodology involves initially defining a set of parameters as input values to HOMER Pro's optimization tool, which generates system configurations with various sizes based on factors like LCOE and TNPC. Input parameters include maximum annual capacity shortage (MACS) and minimum renewable fraction. Maximum annual capacity shortage is the maximum allowable value of the capacity shortage fraction, where the capacity shortage fraction is the ratio of total capacity shortage and total electricity demand [66]. The MACS is the parameter that defines the mini-grid system's reliability and is expressed in percentage (%). The value of MACS depicts how much percentage of the electricity demand is fulfilled by the mini-grid system and how much is being fulfilled by the diesel generators. The minimum renewable fraction represents the minimum share of the generated electricity that is coming from renewable sources and is expressed in % like the MACS. HOMER PRO suggests different system configurations depending on the minimum renewable fraction and different values for MACS.

In the last step, the two off-grid electrification enhancement scenarios are compared based on the abovementioned KPIs. The terminology of break-even grid extension distance is also introduced here to calculate the possible distance up to which the main grid can be extended in the TNPC of the compared scenario [67]. Based on the comparison between the technical and economic viability of the energy systems, a sustainable and reliable energy solution for Tsumkwe and Gam is recommended.

While the methodology thoroughly focuses on the feasibility of the proposed electrification enhancement strategies, a number of assumptions are considered to reach this paper's conclusions and recommendations.

- The economic values of the mini-grid infrastructure prices, including maintenance costs, are based on a literature review and may not be the exact values at the moment of mini-grid implementation. This may affect the LCOE and other economic analyses; however, the authors have used the most relevant sources. Due to the change in the mini-grid operator in the past years, some economic data were not available. For instance, the capital costs were given as approximate values. Likewise, maintenance costs were assumed to be similar to those in 2019, attained from the literature.
- Due to the unavailability of data on the battery performance, the diesel generator running times are assumed to give an insight into battery performance. This assumption would hold true if the diesel generator operation were automated. However, since this is not the case, some discrepancies may arise.
- While data were recorded for over four (4) years, the data used in this paper are mainly those of 2019 and 2021. While these years may not fully reflect the performance of the mini-grid or the off-grid communities best, mean values with other years could not be considered due to either a lot of data gaps or unjustified trends. The demand therefore considered for both communities is assumed to be rather similar to that of the year 2021 in sizing of grid extension capacity and mini-grid extension. No demand forecasts have been performed in this study since this study analyzes the potential of the two enhancement scenarios supplying the current unmet demand.
- Grid extension costs and associated costs were also based on literature reviews and, in some cases, not specific to Namibia. A study previously performed on grid extension in comparison to mini-grid implementation [33] was used as the basis for these costs.

3. Namibia's Off-Grid Communities: Tsumkwe & Gam

To apply the described methodology, this section outlines the system components of the Tsumkwe and Gam mini-grids, along with their operation status. The results and discussions are based on the foundation of the methodology and this paper.

3.1. Tsumkwe Mini-Grid

Tsumkwe is the largest off-grid community in Namibia [68]. In 2012, when the Tsumkwe mini-grid (see Figure 3) was implemented, it was economically unfeasible to extend the national grid to Tsumkwe, and hence a hybrid solar PV-diesel mini-grid was established to electrify Tsumkwe. The mini-grid was officially commissioned by the Namibian Ministry of Mines and Energy in February 2012.



Figure 3. Tsumkwe mini-grid.

Since 2015, Central-North Electricity Distribution Company (Pty) Limited (CENORED) has been responsible for the operation and management of the mini-grid (GPS Coordinates $-19.5985, 20.5037$). The hybrid mini-grid system in Tsumkwe was first established in 2012 with a solar capacity of 202 kWp, which was later expanded by 102 kWp in 2016. The battery system with a capacity of 790 kWh was installed in 2012, later expanded by 1140 kWh in 2014, and by another 1070 kWh in 2016. The diesel generator capacity has been 350 kVA since 2012. Figure 4 shows the evolution of the power plant.

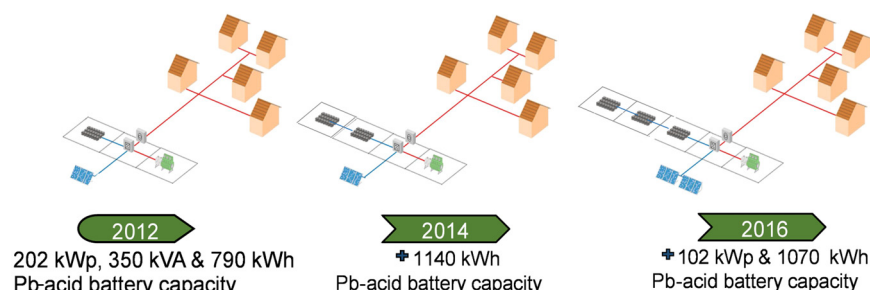


Figure 4. Tsumkwe mini-grid evolution.

The operating condition of the battery system at Tsumkwe was observed to hinder the battery performance. The batteries had degraded over the years because of the high room temperatures affecting the life cycle of the system, since the batteries now operate at high operating temperatures, affecting the achievable number of charging–discharging cycles. As a result, the batteries at the Tsumkwe mini-grid posed as loads rather than storage units.

The Tsumkwe mini-grid serves an off-grid community with approximately 200 households, nine institutions (schools, clinics, and police stations), and supermarkets (amongst other retail businesses). The institutional loads were observed to be the main electricity consumers.

To understand the generation and consumption mismatch, analysis was performed. Figure 5 presents the daily energy consumption and solar production profile of Tsumkwe in 2021. The total annual electricity demand of Tsumkwe was calculated to be 655 MWh, with an average daily demand of 1796 kWh/day. The annual electricity production from PV was 452 MWh, resulting in an annual capacity shortage of 34%. While an annual presentation of solar production and electricity consumption fails to present justification for this mismatch, this 34% shortage could better be explained by daily generation and production trends. As of the afternoon hours, the solar production capacities seemed, on most days, capable of offsetting the instant electricity demand. However, through interviewing the residents and business owners, it was established that demand-side management strategies such as load shedding had been initiated. The evening/nighttime hours, however, mainly showed that the battery capacities could not meet the instant demand, and therefore the irregular introduction of the diesel generator was observed. It could therefore be deduced that this capacity shortage could be due to either insufficient PV production for both instant demand and battery charging or to degradation in battery storage capabilities. Likewise, the electricity production from the diesel generator was summed up and found to be 364 MWh in 2019 and 422 MWh in 2021, highlighting the increased reliance on the diesel generator and/or increased electricity demand, reduction of battery storage capacity, and reduced PV output.

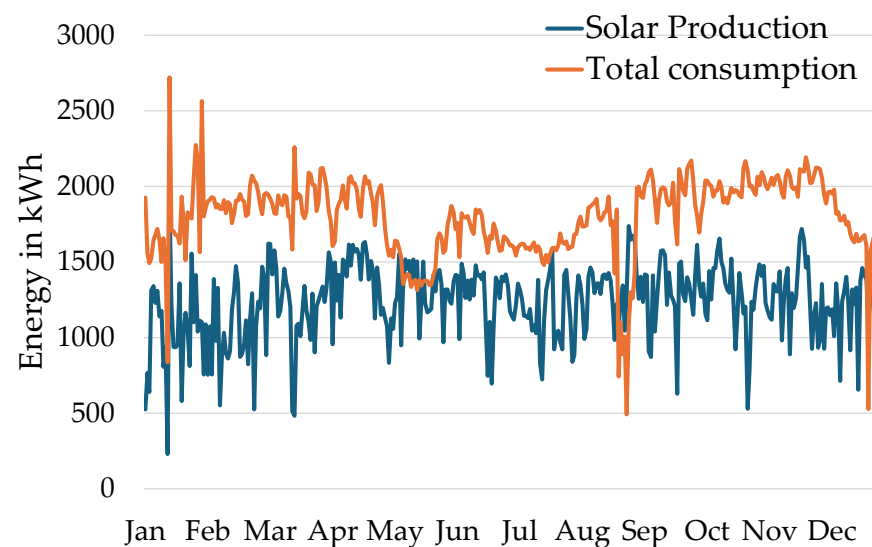


Figure 5. Daily energy profile of Tsumkwe in 2021.

Since there is no monitoring of diesel consumption on site and due to the manual operation of the diesel generator, an estimation for diesel consumption was performed. In general, the specific fuel consumption rate of diesel generators ranges from 0.23 to 0.3 L per kWh [69]. The assumed value throughout this paper is 0.3 L per kWh, which was then multiplied by the generator electricity production in 2021 to estimate the annual diesel fuel consumption [69]. At the calculated 126,600 L/year, and considering each liter emits 2.68 kg of CO₂ [70], the total CO₂ from the consumption of diesel fuel in Tsumkwe was 265,261 kg in 2021.

3.2. Economic Evaluation of System

The economic analysis of mini-grids typically includes the initial cost (C_I), operation and maintenance costs ($C_{O\&M}$), and replacement costs for each component (C_R). C_I includes the component cost and their installation cost, that is, the cost of solar panels and the associated installation cost. It also includes ongoing expenses associated with routine upkeep, monitoring, and servicing to ensure optimal performance and longevity of the components. C_R accounts for the expenditure incurred when replacing components at

the end of their lifetime or due to malfunction. The lifetime (N) of the energy systems considered in this paper is 25 years, and the discount rate (i) is 8% [62]. Considering all these cost factors throughout the lifetime of the mini-grid, an economic evaluation is performed to analyze the system's feasibility. Table 1 shows the cost details for the Tsumkwe mini-grid system. The C_I of PV panels is 1990 €/kWp, and $C_{O\&M}$ is 12.6 €/kWp/year [62]. The lifetime cost of the battery storage system consists of C_I , $C_{O\&M}$, and C_R , which are 325 €/kWh, 20.1 €/kWh/year, and 292 €/kWh, respectively [62]. The converter of the mini-grid system accounts for a smaller share of the total cost than the first two components. The C_I of the converter is 283 €/kWp, $C_{O\&M}$ is 3.87 €/kWp/year, and C_R is 255 €/kWp [62]. The C_I of a diesel generator is 460 €/kW [31]. The cost of diesel fuel is 1.01 €/L [71], and annual consumption is around 126,600 L.

Table 1. Techno-economic details of the Tsumkwe mini-grid.

System	PV (kWp)	Battery (kWh)	TNPC € Million	LCOE €/kWh
Tsumkwe mini-grid system	302	3000	4.68	0.55

The TNPC is calculated by the following formula [72], where N is the lifetime of the system, i is the discount rate, and CF_t is the cash flow in a year between 1 and 25.

$$TNPC = C_I + \sum_{t=1}^N \frac{CF_t}{(1+i)^N} \quad (1)$$

The following equation calculates the LCOE:

$$LCOE_{\text{existing mini-grids}} = \frac{\text{Present value of total cost over the lifetime}}{\text{Present value of electricity generated over the lifetime}} \quad (2)$$

3.3. Gam Mini-Grid

The Gam settlement is the second largest off-grid settlement after Tsumkwe and is located in the Otjozondjupa region [68]. The Gam power plant (see Figure 6) was commissioned in November 2014 and was designed and installed by HopSol. Similar to the Tsumkwe mini-grid, since 2017, CENORED has been responsible for the operation and management of the Gam solar power plant (GPS coordinates: −20.2389, 20.8109).



Figure 6. Gam mini-grid.

The hybrid PV-diesel mini-grid system includes thin-film PV modules with a generation capacity of 292 kWp, a 300 kVA diesel generator, and a lead–acid battery storage capacity of 2600 kWh. Initially, the Gam off-grid community's electricity demand did not

require supply from the diesel generator; however, during the evolution of the community, this has changed. The consumer group of mini-grids is spread over households, businesses, and institutions, for example, schools and clinics.

The energy consumption and generation profile of Gam showed that the mini-grid was previously producing surplus electricity until 2021, when 400 new household connections were connected. This compelled the introduction of a diesel generator in 2020, which previously was not required. Figure 7 illustrates the daily energy consumption and production profile of the Gam mini-grid in 2021. For a few days during the months of January to February and August to October, because of the lack of recorded data, the energy demand values are unknown.

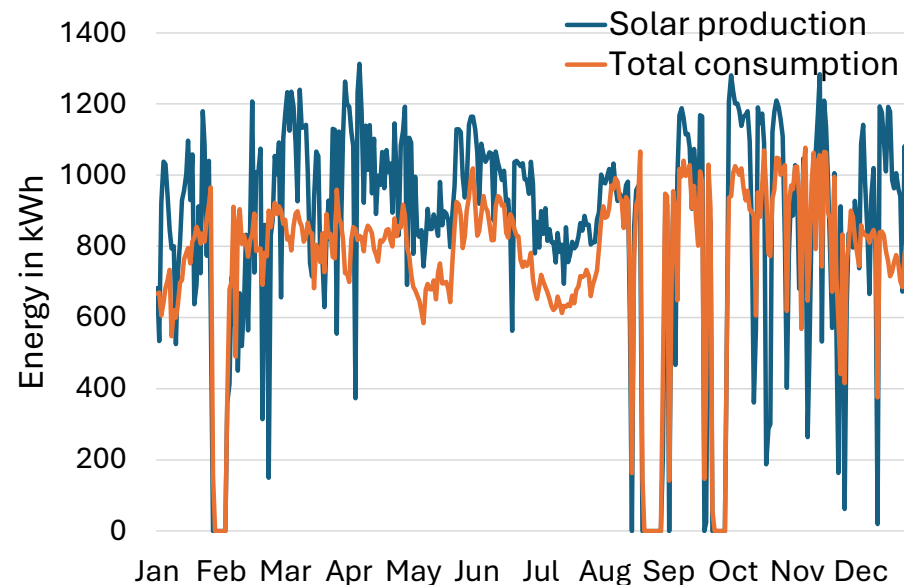


Figure 7. Daily energy profile of Gam in 2021.

In 2021, the recorded total demand of Gam village was 276 MWh, solar generation was 308 MWh, and diesel production contributed around 15 MWh. The average daily energy consumption was around 757 kWh. The total electricity production from the diesel fuel in 2021 was 15,000 kWh, with the total diesel consumption of 4500 L and CO₂ emissions around 12,060 kg.

3.4. Economic Evaluation of the System

Table 2 presents details for the Gam mini-grid system. The TNPC of the Gam mini-grid system is € 3.03 million, and the LCOE is 0.91 €/kWh.

Table 2. Techno-economic details of the Gam mini-grid.

System	PV(kWp)	Battery (kWh)	TNPC € Million	LCOE €/kWh
Gam mini-grid system	292	2600	3.03	0.91

4. Techno-Economic Analysis of Electrification Enhancement Scenarios

Upon establishing the system components and the related challenges for both mini-grids, the analysis of the proposed mini-grid enhancement scenarios is undertaken. Both the technical and economic analyses are carried out for each scenario established in the methodology. This paper explores the techno-economic analysis of these scenarios for both Tsumkwe and Gam and their potential to enhance the power plants in meeting the community's electricity demands.

4.1. Tsumkwe Electrification Enhancement

Grid Extension

As previously explained, the following two sub-scenarios are considered for the national grid extension to Tsumkwe: one with battery storage systems and one without. The nearest national grid point from Tsumkwe is at Mangetti Dune [33], which is 82 km to the west of Tsumkwe. In SSA, the per km cost of grid extension is 30,000–40,000 €/km, depending on the type of terrain [63]. The per km cost considered in this paper is 30,000 €/km, and the $C_{O\&M}$ for extending the grid is 180 €/km/year. The following formula is followed in this thesis to calculate the LCOE in the case of grid extension:

$$LCOE_{\text{grid extension}} = \frac{\text{Total net present cost of extending the grid to measured distance}}{\text{Total energy supply from the national grid}} \quad (3)$$

4.2. Grid Extension with Battery Storage System

In this case, it is assumed that even after extending the main grid to Tsumkwe, the battery storage system is still working as in the designed operating condition/capacity. Hence, the excess electricity from PV would be used to charge the battery storage system. The analysis of Tsumkwe's electricity demand and the generation, an establishment of the mismatched supply, is deduced. To meet this demand, the required electricity supply from the national grid would be 225,392 kWh/year. Using Equation (3), the calculated LCOE is 1.09 €/kWh with the TNPC of € 2.62 million. The mini-grid itself would supply 65% of the demand, and the remaining 35% would be acquired from the grid.

4.3. Grid Extension Without Battery Storage System

Since the battery system at The Tsumkwe mini-grid is already in poor operating condition, this scenario considers a grid extension as a primary backup generator. This assumption rules out the role of the installed battery capacity in Tsumkwe. The electricity generated by the national grid would meet both daytime demand during instances of higher demand than that generated by the mini-grid and nighttime demand. Table 3 presents the techno-economic details for grid extension to Tsumkwe calculated using Equations (1) and (3). The amount of electricity supplied by the national grid in this case is thus higher as it takes on the nighttime demand. The required annual electrical energy supply is thus 391,946 kWh/year. The TNPC remains the same as in the scenarios with battery storage, that is, € 2.62 million. The LCOE is 0.63 €/kWh, around 42% less than the previous scenario due to higher electricity supply from the grid with the same infrastructure as in the previous scenario.

Table 3. Techno-economic details for grid extension to Tsumkwe.

Scenario	Electricity Demand kWh/Year	TNPC € Million	LCOE €/kWh
With batteries	225,392	2.62	1.09
Without batteries	391,946	2.62	0.63

The grid contributes about 59% of the electricity supply, and the mini-grid only contributes about 41% without battery backup.

Mini-Grid Capacity Expansion

The existing mini-grid system in Tsumkwe results in a capacity shortage of 225,392 kWh/year. Using Equation (4) based on [73], the capacity shortage fraction of the mini-grid was calculated and found to be 34%.

$$\text{Capacity shortage fraction} = \frac{\text{Total capacity shortage (kWh/year)}}{\text{Total electrical demand (kWh/year)}} \quad (4)$$

This paper considers the following three cases to mitigate this shortage: 1. 25% capacity shortage, 2. 15% capacity shortage, and 3. 0% capacity shortage, where:

1. 25% capacity shortage implies that the mini-grid can meet 75% of the annual electrical energy demand.
2. 15% capacity shortage implies that the mini-grid can meet 85% of the annual electrical energy demand.
3. 0% capacity shortage implies that the mini-grid can meet the full electrical energy demand.

Depending on these cases, the mini-grid's expanded capacity is optimized in HOMER Pro. According to the abovementioned ranges, the mini-grid would be able to serve the minimum demands ranging from 575 MWh/year to 611 MWh/year compared to 433 MWh/year in 2021. These cases also consider the installed battery capacity to be in full operation condition. Using HOMER Pro, optimal mini-grid extension results (see Table 4) were attained for each case.

Table 4. Tsumkwe mini-grid expansion capacities.

Case	PV Capacity (kWp)	Battery Capacity (kWh)	Additional Generation (kWh/year)	Annual Generation (kWh/Year)
25%	86	-	142,777	575,223
15%	158	-	178,723	611,169
0%	419	1292	225,392	657,838

It is significant to note that only in case 3 (i.e., 0% capacity shortage fraction) is additional battery storage required. In such a case, mini-grid expansion with battery storage expansion, assuming the currently installed is fully operational, would be able to meet the current electrical energy demand of the Tsumkwe off-grid community. For cases 1 and 2, the diesel generator would be sufficient to meet the remaining demand. Based on these findings, economic calculations, including diesel fuel consumption, were calculated (see Table 5).

Table 5. Techno-economic details for mini-grid capacity expansion in Tsumkwe.

Case	TNPC (€ Million)	LCOE (€ Million)	Diesel Use (Liters)	Carbon Emissions (kg)
25%	0.47	0.20	24,300	63,666
15%	0.53	0.23	14,400	37,728
0%	2.20	0.41	-	-

For case 3, there is a considerable surge in TNPC because of the capacity addition of the battery storage system. However, this case presents an advantage in that there is no diesel use and carbon emissions. Especially in the case of remote off-grid communities where diesel attainment can be challenging, this case was necessary to consider.

4.4. Gam Electrification Enhancement

The following sub-sections describe the two scenarios for enhancing the energy supply in Gam as were performed for Tsumkwe.

Grid Extension

The distance from Mangetti Dune to Gam is 143 km. The same approach as for Tsumkwe is followed in the techno-economic analysis. These parameters establish the financial context for evaluating the grid extension's viability and sustainability.

4.5. Grid Extension with Battery Storage System

An analysis in MS Excel revealed that the demand shortage to be met was 11,602 kWh/year. Similarly, an economic analysis was conducted. The limited electricity supply from the grid and the long 143 km distance contribute to the high LCOE. Approximately 95% of the demand is covered by the mini-grid, considering that battery systems will back up the mini-grid during the night and when there are no solar production hours. Table 6 represents the details of the national grid extension scenario to Gam.

Table 6. Techno-economic details for grid extension to Gam.

Scenario	Electricity Demand (kWh/year)	TNPC (€ Million)	LCOE (€/kWh)
With batteries	11,602	4.56	36.86
Without batteries	149,110	4.56	2.87

4.6. Grid Extension Without Battery Storage System

In this sub-scenario, it is assumed that there will be no use of battery storage systems after extending the grid since the community will have a connection to the national grid as a backup. Hence, in this case, the demand considered will be greater than in the previous case. Any excess electricity production from PV would be completely fed into the grid. The electricity supply from the national grid covers the gap during the night and during the day hours when there is no solar production. After performing an analysis, the electricity supply from PV, in this case, is 149,110 kWh/year. Since the electricity supply is higher than in the previous case, while utilizing the same infrastructure, the LCOE is much lower. The grid would supply 54% of the total energy demand, while the rest is from PV production.

Mini-Grid Capacity Expansion

The installed mini-grid system in Gam is currently undersized and can therefore not fulfill the total electricity demand of the community. The possibility of expanding the generation capacity of the mini-grid system was evaluated. For capacity expansion planning of the mini-grid, a set of criteria must be met by the expanded system capacity. The annual electrical energy demand of 276,238 kWh as of 2021 is used in this analysis. In 2021, an energy shortage of 11,602 kWh was observed. This translates to a capacity shortage fraction of 4.2%.

To address this 4.2% capacity shortage within the system, an optimization for the mini-grid capacity expansion was performed in HOMER Pro. Since the value of the current capacity shortage is only 4.2%, this optimization focused only on one case for capacity shortage, that is, 0%, and a minimum renewable fraction of 100%. Table 7 shows the technical details of the expanded system capacity. A 52 kWp mini-grid capacity expansion would meet the demand of the Gam off-grid community without the need to install additional battery capacity. This also indicates that Gam lacks sufficient PV generation to adequately charge the batteries for nighttime demand. This simplifies the expansion since additional battery infrastructure, the most expensive mini-grid component, would not be required.

Table 7. Gam mini-grid expansion capacity.

Case	PV Capacity (kWp)	Battery Capacity (kWh)	Additional Generation (kWh/Year)
0%	52	-	11,602

This case also does not rely on diesel, and therefore no emissions are projected (i.e., within system operation). The economic details are shown in the Table 8 below.

Table 8. Techno-economic details for mini-grid capacity expansion in Gam.

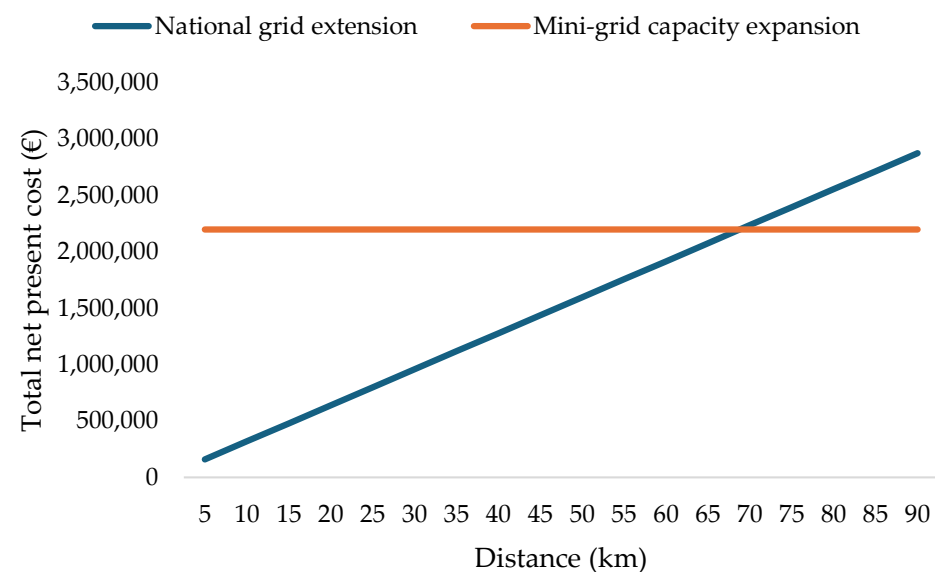
Case	TNPC (€ Million)	LCOE (€/kWh)	Diesel Use (Liters)	Carbon Emissions (kg)
0%	0.17	1.41	-	-

5. Discussion

5.1. Comparison of Scenarios for Tsumkwe

The national grid extension to Tsumkwe is evidently the most expensive option for enhancing electrification. In the case of mini-grid capacity expansion, the cases of 25% and 15% capacity shortage fractions prove a reduction of the LCOE to about half the current value; however, due to the carbon emissions and reliance on diesel, the two options are not the most sustainable, considering that the off-grid community may again have an increased demand in the coming years. The 0% capacity shortage fraction presents a slightly reduced LCOE but presents the cleanest and most sustainable solution, as there is no dependence on the diesel generator. The TNPC, however, then is higher than the other cases.

To fully understand the feasibility of the grid extension to Tsumkwe, the value ‘break-even’ is introduced. This would showcase the distance in km to which grid extension from Mangetti Dune would still be economically viable when compared to mini-grid capacity expansion costs. Beyond the intersection of TNPC of grid extension and mini-grid capacity expansion, grid extension is expensive compared to the other options. In the case of Tsumkwe, the break-even grid extension distance is 68.87 km, which implies that extending the main grid beyond 68.87 km (see Figure 8) is not economically feasible. Since the required distance to extend the grid is 82 km, this rules out grid extension as the best electrification enhancement scenario.

**Figure 8.** Break-even grid extension distance for Tsumkwe.

5.2. Comparison of Scenarios for Gam

In Gam, grid extension is seen to be the most expensive scenario with a TNPC of € 4.56 million. Especially extending the grid with battery storage in Gam, as this means high extension costs for very little generation. In both cases (i.e., with and without the battery storage), the LCOE is too high to be economically feasible for implementation. Among the evaluated scenarios, mini-grid expansion presents the lowest LCOE even though it is still higher than the current LCOE (i.e., 1.41 €/kWh compared to 0.91 €/kWh).

In the case of Gam, the break-even grid extension distance is 5.2 km (see Figure 9), whereas the required distance to extend the grid is 143 km. Extending the national grid to Gam beyond 5.2 km is not economically feasible.

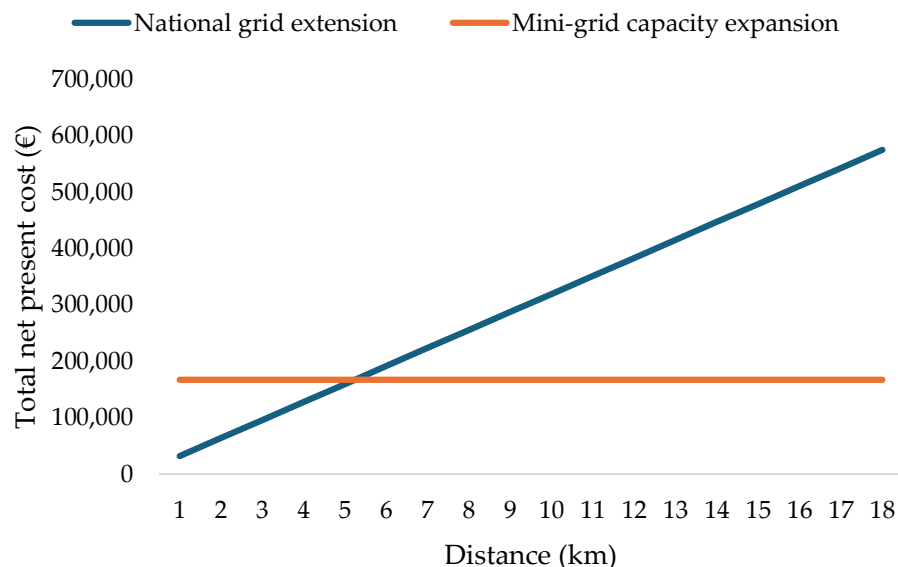


Figure 9. Break-even grid extension distance for Gam.

6. Conclusions

Electrification of off-grid communities in SSA is crucial but often lacks the follow-up upon electricity demand increasing. This is also often a result of policies directed towards electrification with little consideration for sustainability. With time, mini-grids become undersized and fail to meet the communities' electricity demand, and therefore, interventions and innovative strategies are required to enhance the electrification of these communities to eradicate energy poverty. Grid extension and mini-grids, especially solar-based ones, remain the top rural electrification strategies in SSA. This paper, therefore, analyzes the feasibility of grid extension and mini-grid expansion to enhance the electricity supply of existing mini-grids, using Tsumkwe and Gam in Namibia as case studies. It also examines both the technical, environmental, and economic factors, thus ensuring technical sustainability and affordability. The analysis for both Tsumkwe and Gam shows that grid extensions result in a rise in LCOE well above 50% in some considered cases. The determined feasible extension distances are calculated as 68.87 km and 5.2 km for Tsumkwe and Gam, respectively, from Mangetti Dune. This further emphasizes the role of mini-grid capacity expansion. This scenario presents economically feasible options, especially for Tsumkwe, as the attained LCOEs in all three cases (i.e., 25%, 15%, and 0% capacity shortage) are lower than the current LCOE. In the case of Gam, the LCOE remains higher than the current one (1.41 €/kWh versus 0.91 €/kWh). These two communities also present the necessity to assess community needs, existing power systems, and geographical constraints (i.e., in this case, distance to the nearest grid) in determining the most feasible enhancement strategies. While in some cases, therefore, grid extension within the break-even distance boundaries could be feasible, policies need to support the integration of solar mini-grids into the grid and vice versa and likewise prioritize schemes to subsidize electricity costs in off-grid communities. The methodology used can be adapted to any SSAn off-grid community, provided the specific values are changed, respectively, and the key enhancement strategies are also considered in relation to the case study areas. This paper uses assumptions stated in the methodology section due to a lack of data in some cases. Further research using actual data could better compare grid extension with mini-grid expansion in established off-grid communities. Likewise, one may go further to consider policy constraints, financing schemes, and other social aspects that may enhance the sustainability of the enhancement scenarios.

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